SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – Mock Interview

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO5 – Articulation	**Assessment:**	Assessment Tool 3 – Mock Interview
Weightage:	30% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
Student Name:	P. MUNI KARTHIK	**Reg. No.:**	192425061
Section / Batch:	AIML - 2024	**Date:**	

Code of Conduct

I P. MUNI KARTHIK / 192425061 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: P. MUNI KARTHIK

Instructions

- Start the mock interview by entering the provided ChatGPT prompt exactly as given in the Annexure A in the next page. Enter your Name and Register Number when requested.
- Answer all 10 interview questions independently and in your own words using mobile chatgpt voice. Do not copy content from external sources or use AI-generated responses during the interview.
- Provide detailed and meaningful answers with appropriate technical terminology, examples, and justifications wherever applicable. One-line answers may lead to lower scores.
- Complete the interview in a single session. Ensure that all 10 questions are answered before requesting the final evaluation and assessment report.
- Submit the final ChatGPT-generated report printout, including the interview questions, your responses, question-wise rubric evaluation, and the Student Average Final Mark, as proof of completion.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse & Reference Resolution	Understanding discourse, anaphora resolution, reference tracking, and contextual interpretation in text	Q1
Text Coherence & Discourse Structure	Analyzing coherence relations, discourse organization, and maintaining meaningful text flow	Q2
Dialog Acts & Interpretation	Identifying dialog acts, speaker intentions, and conversational meaning	Q3
Dialog and Conversational Agents	Architecture, components, and working of conversational agents/chatbots	Q4
Language Generation Architecture	Stages of Natural Language Generation (NLG), pipeline architecture, and applications	Q5
Surface Realization & Discourse Planning	Transforming semantic representations into grammatically correct sentences and planning discourse	Q6
Machine Translation – Transfer Approach	Translation process using transfer-based methods and practical applications	Q7
Machine Translation – Interlingua Approach	Language-independent representation and multilingual translation framework	Q8
Statistical Approaches in NLP	Statistical language models, probability-based language generation, and machine translation	Q9
Integrated Case Study (Analytical & Problem Solving)	Real-world scenario involving discourse analysis, conversational agents, language generation, and machine translation	Q10

Annexure A – Mock Interview

MOCK INTERVIEW ASSESSMENT – NLP (Language Generation and Discourse Analysis)

Act as an expert Technical Interview Panel for Computer Science and Engineering students.

Interview Domain:
Language Generation and Discourse Analysis

Topics:

- Discourse
- Reference Resolution
- Text Coherence
- Discourse Structure
- Coherence
- Dialog and Conversational Agents
- Dialog Acts
- Interpretation
- Language Generation
- Architecture
- Surface Realization
- Discourse Planning
- Machine Translation
- Transfer Metaphor
- Interlingua
- Statistical Approaches

Interview Instructions:

Step 1:
Ask the student to enter:
Student Name:
Register Number:

Step 2:
Conduct a technical interview consisting of exactly 10 questions.

Rules:

- Ask only one question at a time.
- Wait for the student's response.
- Ask conceptual, analytical, application-based, and problem-solving questions.
- Include at least:
    - 2 questions on Discourse and Coherence
    - 2 questions on Conversational Agents and Dialog Acts
    - 2 questions on Language Generation
    - 2 questions on Machine Translation
    - 2 analytical or scenario-based questions

Step 3:
Store every question and answer.

Step 4:
After Question 10, evaluate the student using the following rubric.

RUBRIC

Technical Knowledge (25)
Level 4:
Demonstrates strong and accurate subject knowledge with in-depth understanding

Level 3:
Shows good knowledge with minor gaps

Level 2:
Basic knowledge with noticeable errors

Level 1:
Lacks fundamental technical knowledge

Problem Solving (25)
Level 4:
Solves problems logically and applies concepts effectively in real scenarios

Level 3:
Applies concepts with moderate accuracy

Level 2:
Limited problem-solving ability; requires guidance

Level 1:
Unable to solve or apply concepts

Analytical Thinking (20)
Level 4:
Analyzes questions critically and provides well-structured logical answers

Level 3:
Provides reasonable analysis with some structure

Level 2:
Superficial or partially structured responses

Level 1:
No analytical thinking demonstrated

Communication (15)
Level 4:
Communicates clearly, confidently and professionally using appropriate terminology

Level 3:
Communicates well with minor hesitation

Level 2:
Communication lacks clarity or confidence

Level 1:
Unable to communicate effectively

Confidence (15)
Level 4:
Displays high confidence, positive attitude and professional behavior throughout

Level 3:
Shows good confidence with minor issues

Level 2:
Low confidence and inconsistent behavior

Level 1:
Lacks confidence and professionalism

MARK CALCULATION

Level 4 = 100% of weightage
Level 3 = 75% of weightage
Level 2 = 50% of weightage
Level 1 = 25% of weightage

Technical Knowledge = 25 marks
Problem Solving = 25 marks
Analytical Thinking = 20 marks
Communication = 15 marks
Confidence = 15 marks

Calculate:
Final Score = Sum of all rubric scores

Step 5:
Generate the final report in the exact format below.

MOCK INTERVIEW ASSESSMENT REPORT

Student Name:
Register Number:

INTERVIEW QUESTIONS AND RESPONSES

Q1:
Question Asked: What is discourse in NLP? Explain how discourse coherence helps a system understand the relationship between sentences in a text.
Student Response:

Q2:
Question Asked: What is reference resolution? Explain how an NLP system can resolve the pronoun in the sentence: “Ravi gave Arun his book.”
Student Response:

Q3:

Question Asked: A chatbot receives the following conversation:

User: “Can I book a flight to Chennai?”

Bot: “Yes. What date would you like to travel?”

User: “Tomorrow.”

Which dialog acts are present, and why are they important?

Student Response:

Q4:

Question Asked: What is a conversational agent? Explain how interpretation and dialog management work together in a chatbot.

Student Response:

Q5:
Question Asked: Explain the main stages of a Natural Language Generation (NLG) system
Student Response:

Q6:

Question Asked: A weather application has the following data: “Temperature = 32°C, condition = sunny, humidity = 60%.” Design a simple NLG process to generate a natural sentence.

Student Response:

Q7:
Question Asked: What are the major approaches to Machine Translation? Explain the difference between transfer-based and interlingua-based translation.
Student Response:

Q8:
Question Asked: Consider the English sentence “The boy is eating an apple.” Explain how an interlingua-based Machine Translation system could translate it into multiple languages.
Student Response:

Q9:
Question Asked: A chatbot answers:
User: “What is the capital of France?”
Bot: “Paris is the capital of France.”
But later the user asks, “What about Germany?” and the bot responds, “Germany is a country.” What went wrong? Analyze the problem using discourse context and dialog management.
Student Response:

Q10:
Question Asked: A Machine Translation system produces the sentence “He went to bank because it was raining,” although the intended meaning is that he went to the river bank. How can discourse analysis and statistical or modern translation approaches help solve this ambiguity?
Student Response:

RUBRIC EVALUATION

Criterion	Weightage	Assigned Level	Marks Awarded
Technical Knowledge	25	Level X	XX
Problem Solving	25	Level X	XX
Analytical Thinking	20	Level X	XX
Communication	15	Level X	XX
Confidence	15	Level X	XX

SUMMARY TABLE

Technical Knowledge (25)	Problem Solving (25)	Analytical Thinking (20)	Communication (15)	Confidence (15)	SubQ Total	Student Avg Final Mark
Level X	Level X	Level X	Level X	Level X	XX.X	XX.X

OVERALL FEEDBACK

Strengths:

- List strengths

Areas for Improvement:

- List improvements

Interview Readiness:

- Excellent / Good / Average / Needs Improvement

End the report after the Interview Readiness section.

After evaluating each answer, do NOT display the marks immediately.

Store the evaluation internally.

After Question 10 is completed:

1. Generate a Question-wise Evaluation Table for all 10 questions.

2. For each question assign:

   - Technical Knowledge Level

   - Problem Solving Level

   - Analytical Thinking Level

   - Communication Level

   - Confidence Level

   - SubQ Total

3. Calculate SubQ Total using:

Level 4 = 100% weightage

Level 3 = 75% weightage

Level 2 = 50% weightage

Level 1 = 25% weightage

4. Calculate Student Avg Final Mark as:

Average of all 10 SubQ Totals

5. Display:

   - Complete interview questions

   - Student answers

   - Question-wise evaluation table

   - Student Avg Final Mark

   - Overall strengths

   - Areas for improvement

   - Interview readiness

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	20	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism

Q. No. / Criteria Level	Problem Understanding	Algorithm Design	Use of Control Structures	Pseudocode Structure	Completeness	Sub. Total	Total %
1	3	3	4	3	3	16	78
2	3	3	4	3	3	16	
3	3	4	3	3	3	16	
4	3	3	3	4	3	16	
5	3	3	4	3	3	16	
6	3	4	3	3	4	17	
7	3	3	3	3	4	16	
8	3	3	3	4	3	16	
9	3	3	4	3	4	17	
10	4	3	3	4	3	16	

Faculty In-charge	Course Coordinator	Program Director
Dr. Kumaragurubaran .T		
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

MOCK INTERVIEW ASSESSMENT REPORT
Student Name: P. MUNI KARTHIK
Register Number: 192425061

INTERVIEW QUESTIONS AND RESPONSES

Q1:

Question Asked:

What is discourse in NLP? Explain how discourse coherence helps a system understand the relationship between sentences in a text.

Student Response:

Discourse refers to a sequence of sentences or utterances that are connected and form a meaningful overall message. Discourse analysis helps NLP systems understand relationships between sentences rather than processing each sentence independently. Discourse coherence ensures that the ideas are logically connected. For example, in "John bought an umbrella. It was raining," the second sentence is related to the first and explains the reason for buying the umbrella.

Q2:

Question Asked:

What is reference resolution? Explain how an NLP system can resolve the pronoun in the sentence: "Ravi gave Arun his book."

Student Response:

Reference resolution is the process of identifying what a word or phrase refers to in a discourse. Pronouns such as "he," "she," "it," and "his" need to be connected to their correct antecedents. In "Ravi gave Arun his book," the pronoun "his" could refer to Ravi or Arun. The system should use grammatical, semantic, and contextual information to determine the most likely reference. Without sufficient context, the sentence can be ambiguous.

Q3:

Question Asked:

A chatbot receives the following conversation:

User: "Can I book a flight to Chennai?"

Bot: "Yes. What date would you like to travel?"

User: "Tomorrow."

Which dialog acts are present, and why are they important?

Student Response:

The first user utterance is a request because the user wants to book a flight. The bot's response is a question or request for information because it asks for the travel date. “Tomorrow” is an answer or inform dialog act because it provides the requested information. Dialog acts help a conversational agent understand the purpose of each utterance and decide what action should be performed next.

Q4:

Question Asked:

What is a conversational agent? Explain how interpretation and dialog management work together in a chatbot.

Student Response:

A conversational agent is a software system that communicates with users using natural language. Interpretation identifies the user's intent, entities, and meaning from the input. Dialog management maintains the conversation state and decides the next action based on the user's input and previous conversation. For example, if the user says “Book a ticket to Chennai,” interpretation identifies the booking intent and destination, while dialog management checks for missing information such as date and passenger count.

Q5:

Question Asked:

Explain the main stages of a Natural Language Generation (NLG) system.

Student Response:

The main stages of NLG are discourse planning, sentence planning, and surface realization. Discourse planning determines what information should be communicated and in what order. Sentence planning converts the selected information into appropriate sentences and decides how ideas should be connected. Surface realization converts the planned representation into grammatically correct natural language. These stages allow a system to generate meaningful and readable text.

Q6:

Question Asked:

A weather application has the following data: “Temperature = 32°C, condition = sunny, humidity = 60%.” Design a simple NLG process to generate a natural sentence.

Student Response:

First, discourse planning selects the important information: temperature, weather condition, and humidity. Sentence planning decides how to combine the information. Surface realization produces the final sentence. For example: “Today, the temperature is 32°C with sunny conditions and 60% humidity.”

Q7:

Question Asked:

What are the major approaches to Machine Translation? Explain the difference between transfer-based and interlingua-based translation.

Student Response:

Major Machine Translation approaches include rule-based, statistical, neural, transfer-based, and interlingua-based approaches. In transfer-based translation, the source language is analyzed and converted into an intermediate representation that is specific to the source and target language pair. A transfer process then converts this representation into the target language. In the interlingua approach, the source sentence is converted into a language-independent meaning representation. The same representation can then be generated in different target languages.

Q8:

Question Asked:

Consider the English sentence “The boy is eating an apple.” Explain how an interlingua-based Machine Translation system could translate it into multiple languages.

Student Response:

The English sentence is first converted into a language-independent semantic representation containing concepts such as boy, eating, and apple, along with their relationships. This interlingua representation does not depend on English grammar. The target-language generation module then converts the representation into the required language. Therefore, the same interlingua can potentially be used to generate translations in Tamil, Hindi, French, or other languages without creating a completely separate source representation for every language pair.

Q9:

Question Asked:

A chatbot answers:

User: “What is the capital of France?”

Bot: “Paris is the capital of France.”

But later the user asks, “What about Germany?” and the bot responds, “Germany is a country.” What went wrong? Analyze the problem using discourse context and dialog management.

Student Response:

The chatbot failed to maintain the discourse context. The second user question “What about Germany?” is related to the previous question about a country's capital. The dialog system should use the previous context to infer that the user is asking for the capital of Germany. The system should resolve the implicit reference and generate an appropriate response such as “Berlin is the capital of Germany.” This demonstrates the importance of discourse history, reference resolution, and dialog-state tracking in conversational agents.

Q10:

Question Asked:

A Machine Translation system produces the sentence “He went to bank because it was raining,” although the intended meaning is that he went to the river bank. How can discourse analysis and statistical or modern translation approaches help solve this ambiguity?

Student Response:

The word “bank” has multiple meanings, so the translation system must use context to select the correct meaning. Discourse analysis can examine surrounding sentences and determine whether the topic is related to a river, finance, or another context. Statistical approaches use probabilities learned from large bilingual corpora to select the most likely translation based on context. Modern neural translation models also use contextual representations and attention mechanisms to consider surrounding words. Therefore, context-aware translation reduces ambiguity and improves translation quality.